POWER PULSE

Error Analysis Important Formulae

Notes by Balaji Sir

1 Basic definitions

1. $$\text{Absolute error} = A_m - A_t, \qquad \text{Correction} = -\text{error}$$

A_m = measured value; A_t = true value

2. $$\varepsilon_r = \frac{A_m - A_t}{A_t}, \qquad \%\text{error} = \varepsilon_r \times 100$$

3. $$\text{Static sensitivity } S = \frac{\Delta\theta_{out}}{\Delta q_{in}}, \qquad \text{Resolution} = \text{smallest detectable change}$$

sensitivity is the reciprocal of the deflection factor

4. $$\text{Limiting error} = \pm\,\%\text{ of FSD} \;\Rightarrow\; \%\text{error at reading } A = \frac{\text{FSD}}{A} \times \%\text{ rating}$$

errors are larger at low readings; use the upper part of the scale

2 Combination of errors (limiting)

5. $$Y = X_1 \pm X_2: \quad \delta Y = \pm(\delta X_1 + \delta X_2)$$

absolute limiting errors add

6. $$Y = X_1 X_2 \text{ or } \frac{X_1}{X_2}: \quad \frac{\delta Y}{Y} = \pm\left(\frac{\delta X_1}{X_1} + \frac{\delta X_2}{X_2}\right)$$

relative errors add

7. $$Y = X^n: \quad \frac{\delta Y}{Y} = \pm n\,\frac{\delta X}{X}$$

3 Statistical analysis

8. $$\bar{x} = \frac{\sum x_i}{n}, \qquad \sigma = \sqrt{\frac{\sum d_i^2}{n-1}}, \qquad d_i = x_i - \bar{x}$$

σ = standard deviation (n − 1 for a small sample)

9. $$P.E. = 0.6745\,\sigma; \qquad \pm1\sigma: 68.3\%,\ \pm2\sigma: 95.4\%,\ \pm3\sigma: 99.7\%$$

probable error and Gaussian confidence limits

10. $$\sigma_Y = \sqrt{\sigma_1^2 + \sigma_2^2} \ \ (Y = X_1 \pm X_2)$$

rms combination of independent random errors